

These graduates can establish startups, join Indian semiconductor companies, or work for global MNCs, strengthening the national talent pool. We also expect this expanding ecosystem to generate many more fabless proposals under Semicon India 2.0. In addition, more than 100 commercial organisations have already gained access to EDA tools and are developing chip architectures while awaiting the next phase of the policy.

**Although C2S has greatly expanded access to chip design tools, complete system-on-chip (SoC) tape-outs remain limited because multi-project wafer (MPW) runs are expensive, often costing ₹800,000-1 million for just 1mm<sup>2</sup>. As India moves towards 28nm manufacturing, is the government considering support for large-scale academic SoC tape-outs on advanced nodes?**

We are actively expanding international partnerships to provide subsidised MPW runs. Through the ChipIN Centre, we already coordinate institutional collaborations for funded tape-outs.

**This year alone, 240 chips have been taped out: 175 at the Semiconductor Laboratory (SCL), Mohali, and 65 through international foundries such as TSMC, GlobalFoundries, and others on advanced node at 28nm and 12nm.** At the same time, we must be pragmatic. Most university projects are undertaken by temporary student teams and are intended primarily for learning rather than commercialisation. Therefore, funding academic tape-outs on leading-edge nodes such as 2nm, 3nm, or 12nm is generally not the best use of public resources, unless the design has clear commercial potential.

For engineering education, mature technology nodes are sufficient to build core design skills. However, the ChipIN Centre will continue negotiating access to advanced international foundries. If a university develops an outstanding SoC with genuine commercial or

strategic value, the government will provide the necessary financial and fabrication support.

**EFY has inspired generations of electronics engineers and entrepreneurs. What message would you like to give future generations considering careers in semiconductors?**

I encourage them to seriously consider careers in electronics and semiconductors. This is a multidisciplinary field that requires expertise not only in electronics but also in physics, chemistry, mathematics, metallurgy, and materials science.

India's semiconductor future is extremely promising. This is becoming a highly rewarding career path, much like computer science. **Taiwan's experience shows that semiconductor engineering salaries eventually matched or exceeded those in software, and I believe India will follow a similar trajectory.**

The sector also offers opportunities to work on world-class research and development (R&D) without leaving the country. In the past, many talented Indian students pursued advanced studies abroad because they saw limited opportunities at home. That is changing. India is building an integrated ecosystem spanning chip design, fabrication, packaging, chemicals, gases, industrial infrastructure, and power electronics, allowing engineers to build global careers while contributing to the country's technological growth.

To support this, the government is introducing specialised curricula and vocational programmes through CBSE, AICTE, the IITs, IISc, and other institutions. The future is extremely bright for anyone who wants to be part of India's semiconductor journey.

**Before we conclude, what gives global investors the confidence to commit to India's semiconductor ecosystem?**

The most important point is the government of India's long-term

commitment to the semiconductor sector. Global companies invest only when they have confidence in policy stability.

Our honourable prime minister, Shri Narendra Modi ji, has repeatedly stated at successive Semicon India summits that semiconductor development is a long-term national strategy, with government support extending over the next 20 to 25 years. This policy certainty, reinforced by funding commitments in the Union Budget, has given investors confidence that Semicon India 2.0, 3.0, 4.0 and beyond will continue seamlessly.

That commitment is reflected not only in policy but also in sustained engagement. The Prime Minister and the Minister for Electronics and Information Technology personally participate in Semicon India events and engage directly with global industry leaders, supported by continuous outreach throughout the year.

India also offers one of the world's most competitive semiconductor incentive frameworks. The implementing agencies operate with transparency, minimise bureaucratic hurdles, and work closely with state governments to ensure projects move forward efficiently. This strong institutional commitment has become one of the key reasons the industry is gaining global confidence.

**Finally, if you were to assess the ISM's journey over the past four-and-a-half years, how would you rate its progress?**

Despite geopolitical shifts, supply chain disruptions, AI-driven demand, and changing global industry dynamics, the project is progressing extremely well. I would rate the journey eight or nine out of ten. **EFY**



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